01/06/26

Bandaging Technique:

Objectives:

- Identify the principles of bandaging and discuss the purpose of the primary, secondary, and tertiary layers of a bandage, including the specific materials used for each.
- Discuss the uses for and characteristics of the following external coaptation methods in small animals and their application technique:
    - Robert Jones bandage and modified Robert Jones bandage
    - Casts and splints
    - Ehmer, Velpeau, and hobbles
    - Bandages for the head, chest, abdomen, tail
- Explain aftercare of bandages, splints, casts, slings, including complications that may occur.

Principles of Bandaging:

- Bandages applied
    - Management of soft tissue
    - Stabilization of bone and joint injuries
    - Debridement
    - Post-op
        - Decrease hemorrhage, edema
        - Eliminate dead space
        - Increase patient comfort
            - Prevents weight bearing
                - Restricts movement
            - Decreases swelling
    - Prevent wound from patient and environment

Supplies:

- Telfa-pads
- Cast padding
- Rolled cotton
- Stretch gauze
- Porous tape
- Vetwrap
- Elastikon

Bandage 3-layers:

- Stir-ups
- Primary
    - Direct contact
    - Adherent vs non-adherent
    - Occlusive, semi-occlusive, non-occlusive
- Secondary

        - Padding
        - Stretch gauze
- Tertiary protective
    - Holds and protects other two layers

<u>Primary Layer- Bandage Dressings:</u>

- Sugar dressing
    - High osmolarity
        - Draws fluid out of the wound
    - Inhibit bacteria growth
    - Aids in debridement while preserving tissue
- Honey dressing
    - Hydrogen peroxide production
    - Low pH accelerates healing
    - Unpasturized
    - Manuka honey
- Antibiotics
    - Sprays
    - Creams

<u>Wound Debridement: Types of Bandages:</u>

- Dry to dry, wet to dry, wet to wet
- Adherent primary layer
    - Debris becomes incorporated into the bandage
    - Debris removed when bandage is removed
    - Wet to dry
        - Saline applied to wide-mesh gauze. Removed after it's allowed to dry
        - Painful, less desiccation
    - Dry to dry
        - Wide-mesh removed after primary layer absorbed fluid
        - Painful to remove, excellent tissue debridement
    - Wet to wet
        - Removed before it's allowed to dry
        - Tissue damage due to moisture

<u>Bandage Application in Small Animals:</u>

- Distal limb bandages
    - Wound debridement
    - Robert Jones
    - Modified Robert Jones
        - Most common in small animals
    - Tie-over
    - Splints
    - Slings

    - Casts

Modified Robert Jones:

- Wound bandage
- Basis for all other bandages/splints
- Virtually any area

Casts and Splints:

- Used:
    - To provide stabilization and support for the joint injuries
    - For temporary support until surgery
    - As a means for definitive treatments
    - As an adjunct after surgical stabilization
- Only for injuries below elbow or stifle
    - Quick splint
    - Spoon splint
    - Sam splint
    - Modified Thomas (Schroder Thomas)

Splints:

- Mason-Meta Splint (Spoon Splint)
    - Supports distal limb injuries
- Spica Splint (Quick splint)
    - Maintains forelimb or pelvic limb in extension
    - Most commonly used to keep elbow immobilized
    - Support of other types of fixation

Bandage/Splint Application for Horses:

- Bandaging
    - Wounds
    - Trailering
    - Ace bandaging or trach wrap
        - Wrap tape around “barber pole”: fashion
    - Splinting
        - Extensor or flexor tendon lacerations
        - Flexure deformities in foals

Casts:

- Plaster of Paris
    - 5-10 minutes to set up
    - Gets hot during set-up
    - Very heavy and susceptible to moisture
- Fiberglass (e.g. Hexcelite, Vet-Cast)
    - Lightweight, rigid, waterproof

        - More expensive

Cast Application:

- Gloves
- Stockinette
- Modified Robert Jones
- Additional padding
- Time sensitive placement of cast material
    - Moisten cast material
- Incorporate stockinette into cast

Wound Management Aftercare:

- Timing
    - Risks of prolonged immobilization
        - Bandage changes
    - Limiting exercise
        - Monitoring distal limb bandages
            - Protective devices
                - Plastic bags; E-collar
            - Complications
        - Client education

Wounds and Management:

Wound Healing:

- Lecture Objectives
    - Discuss how wounds heal and the influences on that process
    - Ability to actively monitor wound healing
    - Understanding and choosing the correct management for various types of wounds

Terms to Know:

- Abrasion- scrape, takes off the epidermis
- Collagen- protein that has elasticity
- Contraction- things coming together
- Dead space- any space that is not properly secured
- Debridement- removal of dead skin and bacteria from wound
- Degloving injury- removal of skin
- Decubitus ulcer- bed sores
- Epithelialization- thickening of the skin- skin healing on the top of the wound
- Granulation tissue- pink, shiny tissue
- Fibroblast
- “Golden period”
- Inflammation
- Laceration

- Maturation phase
- Primary closure
- Second intention healing
- Proliferative phase

Integument Review:

- What are the layers of the skin?
    - Epidermis
        - Langerhans Cells- macrophages
        - Regulate immune response to antigens
    - Dermis
        - Nourishment to epidermis
        - Macrophages
        - Lysosomal enzymes (complete destruction)
    - Hypodermis
- What layers are most important in regards to wound healing?

Wound Healing Intro:

- Phases-
    - Hemostasis/inflammation
    - Proliferation
- Cells
    - Neutrophils
    - Macrophages
    - Fibroblasts

Healing- Hemostasis and Inflammation:

- Vasoconstriction
- Vasodilation
    - Damaged cells release mediators
    - Capillary diameter and permeability
        - Warm, red, swollen
        - Neutrophils
        - Platelets, clotting factors, and fibrinogen
- 6-8 hours
- Clot
- Scab

Healing- Inflammation-Debridement:

- Removal foreign material and necrotic tissue
- Neutrophils
    - Minutes to hours
    - Phagocytosis
    - Enzyme release

        - Pus formation
- Macrophages
    - Days 0-5
    - Phagocytosis
    - Attract fibroblasts

<u>Healing- Proliferation & Repair:</u>

- Granulation tissue
    - This is a good thing!
    - Red, shiny, moist
    - Bleeds easily
    - Not painful
    - Not prone to infection
        - Protects the wound from invaders
        - Fills wounds with tissue
        - Replaces necrotic tissue
    - Disturbing to clients
- Begins day 3-5
- Fibroblasts
    - Collagen synthesis
- Capillaries
    - Oxygen
    - Nutrients
    - Moisture
    - Warmth
- Macrophages
    - Continued debridement
- Epithelium
    - Begins day 4-5
    - Proliferation
        - Up to 5 mm from edge of wound
    - Migration
        - Beneath scab in apposed wounds
        - Over granulation tissue in open wounds
- Wound contraction
    - Begins day 5-9
    - Contraction of myofibroblasts in granulation tissue
    - Wound margins pulled together
        - Limited by skin tension
    - A major part of healing in most animals, but not humans

<u>Healing-Maturation:</u>

- Scar remodeling
    - Day 21 to years

        - Collagen realigns, links and strengthens
        - 80-85% of normal
        - Further contraction
            - Pink, raised to white, hard and flat

What does A Wound Need to Heal?

- Oxygen
- Nutrients
- Moisture
- Warmth

Patient Factors That Influence Wound Healing:

- Nutritional status
    - Is patient eating
- Origin of wound
- Corticosteroid administration
- Radiation therapy
- Orthopedic or neurologic problems
- Age
- Treatment
    - What kind of meds has the client decided to use?
- Hydration status of patient
- Concurrent disease:
    - Diabetes, Cushing's FeLV, Cancer
- Obesity
- Location
    - Blood supply may vary with location
- Origin
- Depth
- Contamination or infection
- Location of wound
- Moist environment is preferred
- Irritation
    - Trauma, motion, chemicals, infection
- Improper dressing and bandaging

Enemies of Wound Healing:

- Tension
- Pressure
- Movement
- Tissue trauma
- Contamination
- Infection
- Excessive granulation tissue (proud flesh)

Steps to Wound Care:

- Address pain
  - Anesthesia is often necessary to achieve goals
- Wash your hands
- Wear gloves
- K-Y trick
  - Trap hair before clipping, water soluble, sterile, use carefully
- Clip wound
- Gently clean surrounding skin
- Lavage-
  - LRS, but water is ok, Ideal psi (8-12), using 35 ml syringe and 18-19g needle
  - Brings down level of bacteria to a safe level
- Some add antiseptics
  - Very, very dilute. Cytotoxic. (1/40 chlorhex or 1/10 betadine)
- Cover wound with appropriate bandaging technique, if possible
- Debridement
- Explore wound- a word about bite wounds- dogs shake and tear, cats puncture
- Treat dead space- Why?- fluid accumulation

Classification of Wounds:

- Clean
  - Surgical incisions
- Clean-contaminated
  - Surgery into GI, respiratory, or urinary system
- Contaminated
  - Open wound trauma
  - Surgical spillage
- Dirty
  - Gross contamination
  - Active infection

Types of Wounds:

- Abrasions
- Lacerations
- Degloving injuries
- Bite wounds
- Burns
- Decubitus ulcers
  - Pressure sores

Types of Wounds- Abrasion vs Laceration: Depth:

- Abrasion
  - Loss of epidermis and part of dermis

- Laceration
    - Irregular wound caused by tearing of tissue with variable damage to superficial and underlying structures

<u>Bite Wounds:</u>

- Dog vs cat
- Crushing damage to the skin, SQ tissues and internal organs
    - May not be obvious on PE
    - Usually needs to be surgically explored
- Penetrating wounds

<u>De-Gloving Injury:</u>

- Common in dogs that jump off or fall out of a moving vehicle and are dragged along the pavement, also seen in fan belt cars or animals caught by fence, as this one
- Slow to heal, may need grafting, dedicated attention to cleanliness and preservation of healthy tissue

<u>Burns:</u>

- 1st degree
    - Epidermis ONLY
    - Erythema and swelling
    - Sunburn, clipper burn, chemical burns
- 2nd degree
    - Into dermis, blistering possible
    - Moist, red, swollen
    - Very painful
- 3rd degree
    - Full thickness injury
    - Deep pain only
    - Must debride surgically
- Management
    - Pain management
    - Fluid therapy
        - Fluid, electrolyte and protein loss
    - Antibiotics
        - Topical and systemic
        - Strong and broad spectrum
    - Nutritional support
        - Demand may be double or more
        - Stomach tubes if necessary
    - Husbandry
        - Use sterile gloves and clean towels when working with wounds, especially burns
            - Dawn

- ICU

Gun Shot:

- Damage is a result of the concussive force
  - F=ma: size and speed of projectile
- Must be surgically explored to determine the extent of damage
- High potential for pneumothorax or hemothorax

Wound Drainage:

- Dead space
- Passive drain
  - Capillary action
  - Penrose drain
- Active drain
  - Closed-suction
  - Vacuum action
  - Rigid fenestrated drain

Wound Debridement:

- Wound debridement
  - Removing diseases/dead tissue to promote healing of healthy tissue
  - Surgical, enzymatic, mechanical
  - Staged

Types of Intention:

- Primary
  - Uncomplicated healing of non-infected, well-approximated wound is defined as primary healing
    - Surgical
- Secondary
  - Wound healing course is disrupted by injection, dehiscence, hypoxia or immune dysfunction (no closing- leave it open!!!)
  - Granulation tissue formation and epithelization
    - More susceptible to infection and poor healing
- Tertiary/ third intention
  - Delayed primary wound healing after 4-6 days
  - Secondary intention is interrupted and wound is closed
    - After granulation tissue is formed

Types of Wound Closures:

- Primary closure
  - Wound close within 6-8 hrs (golden period)
  - Primary intention healing
- Delayed primary closure
  - Open wound for 2-3 days

        - Closure of wound before granulation tissue formation
            - Cleaning required, patient unstable
    - Secondary closure
        - Closure of wound after development of granulation tissue
            - Third intention healing
    - Second intention healing
        - Wound left to heal without closure

Wound Closure:

- Primary closure with primary intention
    - Suturing of wound
        - Less than 6-8 hrs
- Delayed primary closure
    - Some contamination
    - Open wound for 2-3 days
- Secondary closure (3rd intention healing)
    - Infected necrotic wounds
    - Forms healthy granulation tissue then sutured closed
- Second intention healing
    - Wound left to heal without closure

Wound Infection

- Signs
- Determining severity
- Choice of antimicrobial
    - Oral vs IV
    - Type of pathogen
- Resistant organisms
    - Culture and testing
- Topical antimicrobials

Wound Problems:

- Infection
    - Inflammation
    - Discharge
    - Odor
- Devitalized tissue
    - Ischemia- lack of blood flow
    - Discoloration
    - Dry, hard “eschar”- necrosis
- Fluid accumulation
    - Seroma, hematoma, abscess
    - Very prone to infection
    - Requires drainage and evaluation
- Dehiscence

        - Wound repair breaks down and reopens
        - Peaks at 4-6 days, weakest point in healing
        - Infection, tension, motion, poor circulation, etc

01/20/25

Dermatology Clinical Nursing Skills:

Dermatology Objectives:

- Physical exam
    - Normal
    - Abnormal
- Terminology
- Nursing assessments
- Brief disease talk
- Common diagnostics
    - Clinical nursing skills you should know how to do!
- Client education

Normal Skin Anatomy:

- Largest organ
- Covers and protects
    - Barrier function
        - Bacterial, UV, trauma
    - Many structures
        - Epidermis, hair, hooves, horns, claws, gland
    - Contagious with mucous membranes
        - Skin has a superpower! It can regenerate and heal

Functions of Integument:

- Prevents desiccation
- Reduces injury
- Immunity
- Temperature control
- Excretes water, salt, and waste
- Sensory organ
- Stores nutrients, can synthesize Vitamin D
    - Species dependent

Physical Examination:

- Look and feel and smell everywhere!
    - Coat
    - Skin
    - Inguinal
    - Axillary
    - Ears

    - Nails/nail beds
    - Interdigital
    - Nose, pads
    - Anal scs
    - Mucocutaneous junctions
    - Tails
    - Rectal
    - Lymph nodes

Nursing Assessments:

- Can you think of some findings you might use to describe changes in the skin, coat, and other parts of the dermatologic system?

HIstory Questions to Ask:

- When did the problem begin?
- Is it a seasonal problem?
- What is the day-to-day pattern?
- To what is the animal exposed? Anything new?
- Get the family history- is it a genetic problem seen in sire, dam, siblings?
- Any other household pets affected or human companions?
- Nutritional intake
- Housing

Ectoparasites:

- Fleas
    - Flea diagnostics: flea combing!!
    - Where to flea comb?
        - Consider the “flea highways”: Lumbosacral region, under the chin
- Mites
    - *Sarcoptes scabiei*
    - *Notodres cati*
    - *Demodex folliculorum*
    - *Cheyletiella spp.*
    - *Anoplura*
    - *Malophaga*
    - *Trichodetes spp.*

Notoedres cati= Feline Scabies:

- Zoonotic, but not limited

Cheyletiella spp.:

- Known as “walking dandruff” = skin scales are carried by mites
    - Zoonotic!!!

Food Allergies:

- Not as common as clients think, but do exist.
- Hard data tough to find, possibly <5% of all allergies
- Requires a 12 week food trial of a novel protein diet- and reintroduction of original food- to diagnose!

Pyoderma:

- Superficial or deep
- Bacterial infection in the skin
- Multiple causes!!
- Any skin disease can result in a pyoderma
- Best treated with oral and topical antimicrobial therapy

Pemphigus Foliaceus:

- Immune-mediated disease
- Commonly seen as crusts at the pinnal margins, nasal planum, digital pads

Dermatophytosis= “Ringworm”:

- Microsporum canis
- Microsporum gypsum
- Trichophyton spp.
- This is a fungal disease!!! Difficult to treat.
- Zoonotic!!

Ecto-Ectoparasites:

- Fleas
  - Ctenocephalides
- Lice
  - Anaplura- sucking
  - Mallophaga- biting
    - Trichodectes canis
    - Felicola subrostratus
- Ticks
  - A whole bunch

Diagnostic Tests:

- Skin scraping
  - Mites
- Flea comb
  - Fleas, lice, mites, chiggers, ticks, epilation, scale
- Cytology
  - Cells, bacteria, yeast
- DTM
- Culture

        - Pustules, wounds
    - Wood's lamp
        - 70% dermatophytes fluoresce

How to Run Various Skin Tests:

- Flea comb
- Impression smear/cytology
- FNA
- Biopsy
- Culture
- Wood's lamp
- Skin scraping
- DTM
- Materials
    - Slides, mineral oil, blade, stains, Q-tips, culture media, Wood's lamp, needles, cellophane tape?
- Steps
- Interpretation
- Role of technician

Performing a Skin Scraping:

- Dull blade
    - Dip in mineral oil
- Pinch skin
    - Push mites from follicles
- Deep skin scraping until you see capillary bleeding
- Describe pattern: local or generalized
- Sensitive area
    - Pluck hairs and look at follicles for demodex

Microscope Work:

- Reading a slide
- Staining
- What are we looking for?
    - Mites
    - Yeast
    - Bacteria
        - Rods, spirochetes, cocci, gram -, gram +

Treatment:

- Address patient comfort
- PPE
    - Zoonotic
- Work-up before clipping and cleaning

        - Diagnostics
- Clip and clean
    - Clip with the grain, clean with the grain
- Medicated bath +/-
- Topical medication
- Oral medication
- Client education

<u>Bathing- Types of Shampoo:</u>

- Medicated bathing
    - Anti-fungal
    - Anti-parasitic
    - Allergens
    - Bacterial/yeast infections
- Regular shampoos- recommend oatmeal base

<u>Bathing- Full Bath- Procedure:</u>

- Ears
    - Cotton balls
- Eyes
    - Eye lubrication
    - Mineral oil
- Shampoo
    - Wet coat thoroughly then apply small amounts of shampoo
        - Head to tail
- Rinse
    - Axillary, inguinal regions
- Dry
    - Towel, dryer

<u>Client Education:</u>

- 80% of pet owners admit to not carrying out all the shampoo treatments
    - Frequency of bathing
    - Stay on?
    - Areas to avoid
    - Procedure
    - When to discontinue
    - Alternatives

<u>What is the Role of the RVT?</u>

- History
- Skin diagnostics
- Initial therapy
- Advocating for patient comfort

- Client education/client support
    - Frequency, tips, etc.

01/22/26

Grooming:

- Nail trims
- Bathing
    - Procedures
    - Types of shampoo
- Clipping
- Client education

Terminology:

- Anal sac disease terms- impaction, abscess
- Anal sacculectomy
- Scooting
- Enema
    - Catheter tip syringe
- Lube, lube, lube
- Megacolon

Viewing the Perineum:

- Knowing normals
- Temperature
- Collecting fecal sample
- Alopecia, erythema, discharge, swelling
- Lubrication will not interfere with fecal results

Expressing Anal Sacs:

- Indications
    - Scooting, licking, turning, odor
- Location
    - Canine 4 and 8 o'clock positions
    - Feline 3 and 9 o'clock positions
- Supplies
    - Gloves, lubricant, paper towel or gauze, baby wipes, chlorhexidine

Otic Clinical Nursing Skills

Lecture Terms:

- Pinna
- Vertical canal
- Horizontal canal
- Tympanic membrane
- Otitis

- Cytology
- Otodectes cynotis

Lecture Objectives:

- What does the external ear do?
- What are the functions of the ear?
- Division of external, middle, and inner ear
- How to examine the ear for physical exam
- Basic diagnostic tests for ear pathology
- Basic ear cytology
- How to restrain and clean ears
- Common aural disease pathology
- Common aural surgical procedures

Basic Structures:

- External ear:
    - Pinna- “ear flap” = many shapes in canine!
    - Funnels sound waves into the canal
    - Mobile- turn in the direction of sound
    - External auditory canal- soft tube that conducts sound waves to the ™
- Not a straight tube-
    - Outer portion is vertical and inner portion is horizontal
- Middle ear
    - Air-filled cavity within the temporal bone
    - Between tympanic membrane (™) and cochlea
- Outer ear= pinna + external auditory canal
- Tympanic membrane= boundary between outer and middle ear
- Middle ear= ossicles + middle ear cavity
- Inner ear= cochlea + vestibule + semicircular canals

Examination of the Ear:

- Must include inner and outer structures/surfaces
- A thorough examination includes 3 independent parts:
    - Examination of the pinna and surrounding area/ skin
    - Examination/assessment of the tympanic membrane (requires otoscope!)
    - Otoscopic examination of the ear canal in general
- Some patients will require sedation
- Different tests/tools will be utilized
    - Otoscope
    - Cytology
    - Skin scraping
    - Mite check
- Must know the anatomy of the ear to do a thorough exam

01/27/26

Examination of the Pinna:

- Remember your dermatology-
    - Pustules on the pinna can indicate infectious disease (bacterial infection?)
    - Papules can indicate parasitic infection or disease (fleas, mites?)
    - Erythema can indicate sarcoptic mange or dermatophytosis, or topical medication allergy
    - Vasculitis can be a result of ulceration, necrosis
    - Lichenification can indicate chronic dermatitis

Continuing Examination of the Pinna:

- Color of pinna
- Exudate
- Crusting or thickening
- Smell
- Signs of pain
- Signs of self trauma

Otoscope:

- Insert cone into the ventral ear canal
- Identify lumen in the center of the eyepiece advance further
- Ear canal will narrow and bend
    - Facilitate advancement of the cone by gently pulling the pinna outward and down
- After the bend, you are in the horizontal canal
    - Visualize the tympanic membrane

Cytology:

- Simple and quick in-house test
- Gently insert cotton-tip applicator ("Q-tip") into the horizontal canal
- Roll the sample onto a microscope slide and stain
    - Heat fix the sample
    - Stain with Wright's stain
- Examine under a microscope
    - Ensure you check the slide in its entirety

Ear Cytology:

- Cocci and bacillus (bacteria)
- Infection
    - Neutrophils with intracellular bacteria
- Yeast: Malassezia

Exam for Ectoparasites:

- Patients will be pruritic with alopecia
- Sample taken similarly to cytology

    - Can also do a deep skin scraping
- Usually the debris in the ear looks a bit differently parasites are suspected
- Think “coffee grounds” in a cat ear= often ear mites

Otitis Externa/Media:

- Clinical signs:
    - Inflamed, painful ears with an odor, discharge, head shake, head tilt, rubbing and scratching
    - Genetic predisposition: Some breeds (Spaniels) produce excess cerumen and epithelial cells in ear canal
- Diagnosis: PRIOR TO TREATMENT
    - Otoscopic and microscopic exam
    - Culture for bacteria and yeast (fungal)
- Treatment:
    - Control any predisposing factors
    - Pluck excess ear hairs
    - Clean/flush the ear, then medicate properly

Ear Mites:

- *Otodectes Cynotis*
    - Tresaderm ointment
    - Ivermectin injection
    - Mitacide
    - Selamectin (Revolution)
    - “Coffee grounds”

Polyps/Growths in the Ear:

- Polyps: Type of benign tumor. They grow from the ear canal, have a stalk and can grow significantly in size
- Can cause persistent ear infections- esp in cats
- Signs: scratching ear, shaking head, head tilt, loss of balance

Aural Hematoma:

- Mainly seen in dogs, rare in cats.
- Caused by the rupture of blood vessels between the skin and the cartilage
- Usually secondary to some ear irritation
- Treatments:
    - Aspirate with the use of a syringe
    - Place drains
    - Definitive surgical treatment
        - Suture removal in 30 days
    - Must correct underlying cause or will recur

Tympanic Membrane Rupture:

- Painful!!
- Causes:
    - Trauma, infection, sudden severe change in atmospheric pressure, loud noises, foreign bodies
- Treatment:
    - Deep ear cleaning (+/- sedation or anesthesia), limited topical medications, systemic antibiotics
    - Long time frame to heal (4-8 weeks)

Foreign Body:

- FOXTAILS!!! Grass, seeds
- Signs:
    - Same as with any other symptom of ear irritation
- Treatments:
    - Removal of foreign object
    - Can cause ear infections, aural hematomas, perforated tympanic membrane… need to be ready to treat all outcomes

To Pluck or Not to Pluck:

- Debate that goes on about whether to pluck the ear in an ear canal
- Most groomers will do this- not all breeds have an overgrowth of hair in their ears
- Why would you?
    - Think about the shape of the ear canal, how dark and moist it is..
- How it is done:
    - Simple restraint is usually enough
    - Curved hemostat- careful to only grab hair!
    - Some places use powder to dry the hair for an easier grab
    - Twist and PULL!! Also…twist, twist, twist, method

TECABO:

- A surgical intervention when a patient has recurrent ear infections
- “Total Ear Canal Ablation and Bulla Osteotomy”- removal of ear canal and tympanic bulla (inner ear)

TECA:

- Total Ear Canal Ablation
- Most radical tx
- Completely remove and close the ear canal
- Neoplasia
- Severe, chronic inflammation
- All other tx has failed
- Usually results in deafness in that ear

Ear Cleaning:

- Check the tympanic membrane prior to cleaning!!!
    - If ruptured, use sterile saline only!!
- Sedation or anesthesia
- Variety of cleaning agents commercially available- do not use Isopropyl Alcohol
- Once a cleaner has been determined:
    - Clean “as far as you can see” with cotton tip applicators “Q-tip” and cotton balls. No gauze
    - Do not advance any farther than you can see to avoid pushing debris further into the ear
    - Cleaning of the interior ear may involve a bulb syringe (or irrigation device)
        - Be cautious with force- can rupture the tympanic membrane

02/03/26

Blood Pressure and ECG:

Cardiac Cycle:

- Coordinated sequence of events that occurs during one heartbeat
    - Contraction and relaxation of the atria and ventricles
    - Movement of the electrical impulses through the conduction system
    - Opening and closing of the valves
    - Filling and opening of the heart’s chambers

Cardiac Auscultation:

- Heartbeat sound= “lub dub”
- “Lub”= closure of the atrioventricular valves
    - Atria are emptied; blood is loaded into ventricle
- “Dub”= closure of aortic and pulmonary valves
    - Blood has exited the ventricles
- Sinoatrial Node (SA)
    - Primary pacemaker
- Atrioventricular (A-V) node
    - Secondary pacemaker
- Bundle of His
- Left and right bundle branches
- Purkinjie fibers
    - Tertiary pacemaker
- **Depolarization**: release of electrical impulse which cause the cardiac muscles to contract
- **Repolarization:** Recharging of electrical impulse in preparation for depolarization- relaxation of the heart

Depolarization and Repolarization:

- Cell at rest more positive ions outside than in
    - Negative inside the cell (potassium) and positive outside the cell (sodium)- polarized cell

- Cell receives signal- sodium channel opens
    - Na enters the cell changes polarity- depolarization (+cell, -outside)
- Potassium channel opens
    - Potassium leaves cell changes polarity- repolarization (-cell, +outside)
- Sodium potassium pump
    - Uses ATP to put everything back

Cardiac Cycle Review:

- There are a few things that happen in synchronism for every heartbeat to allow for appropriate perfusion to the body
- Mechanical and electrical
    - P wave
        - Atrial depolarization
        - Atrial contraction
    - QRS complex- systole- pulse felt
        - Ventricular depolarization
        - Ventricular contraction
        - Atrial repolarization
        - Atrial relaxation
        - AV valves close (lub)
        - Semilunar valves open
    - T wave
        - Ventricular repolarization
        - Ventricular relaxation
        - AV valves open
        - Semilunar valves close (dub)

Electrocardiogram (ECG):

- Provides a way to measure the movement of the electrical impulse through the heart during the cardiac cycle
- P wave
    - Depolarization of the atria
    - Atrial systole
- QRS complex
    - Depolarization of the ventricles
    - Ventricular systole
    - Repolarization of the atria occurs but is not seen through the large wave created by the more muscular ventricles
- T wave
    - Repolarization of the ventricles
    - Relaxation/regrouping of the ventricles
    - Ventricular diastole

Normal ECG- Lead II:

- PR interval
    - Onset of P wave to start of QRS complex
- PQ Segment
    - End of P wave and start of QRS complex
- ST Segment
    - End of S wave and beginning of T wave
        - Interval between ventricular depolarization and repolarization
- QT interval
    - Start of Q wave to the end of T wave
        - Time taken for ventricular depolarization and repolarization
            - Time takes for ventricular depolarization and repolarization
                - Period of ventricular contraction to relaxation

Indications For the Electrocardiogram:

- Arrhythmia
- History of syncope or episodic weakness
- Cardiac chamber enlargement
- Pericardial effusion
- Pericardiocentesis
- Monitoring drug therapy
- Monitoring during anesthesia

Tracing Leads:

- I- right arm to the left arm
- II- Right arm to the left leg
- III-Left arm to the left leg
- aVR- right arm to a point midway between the left arm and left leg
- aVL- left arm to a point midway between the right arm and left leg
- aVF- Left leg to a point midway between the left arm and right arm

02/05/26

Acquisition of the Electrocardiogram:

- Right lateral recumbency
- White electrode- right forelimb (RA)
- Black electrode- left forelimb (LA)
- Green electrode- right hindlimb (RL)
- Red electrode- left hindlimb (LL)

Electrocardiogram:

- Theory
    - Measures the electrical depolarization of the heart
- What it tells you
    - Heart rate

    - Heart rhythm
    - Impulse conduction
- What can you infer
    - Chamber enlargement
    - Myocardial disease
    - Ischmia- lack of O2 causing damage to the muscle tissue
    - Electrolyte imbalances
    - Pericardial disease-fluid in the sac surrounding the heart, thickening of the sac
- What does it NOT tell you
    - Strength or presence of contractions
    - Cardiac output- how much blood is being pumped
    - Function of the chambers of valves
    - Size and shape of heart
    - If your heart is beating normally

<u>Performing an ECG</u>

- Padding or mat if steel table
- Conducting gel, or alcohol (avoid if defibrillation is a possibility)
- Alligator clip or pad (must clip)
- Right lateral recumbency is ideal. (Large animals stands, cats can be sternal, if dyspneic-sternal is ok)
- Challenges: panting, noisy room, other electronics

<u>What Can Complicate an ECG Tracing?:</u>

- Movement!
    - Shivering patient
    - Respirations (panting=movement)
- Other electrical activity/devices near the patient (unplug!)
- Not enough gel or alcohol on alligator clips
- Limbs not perpendicular
- Electrodes in the wrong location (should be just distal to the elbows and stifles!)
- Patient is in wrong position

<u>Heart Rhythm-Normal:</u>

- Normal Sinus Arrhythmia (NSA)
    - Rate changes with breathing
    - Caused by increased Vagal tone
        - Stimulates the parasympathetic nervous system (“rest and digest”)
        - Lowers heart rate
        - Senses thoracic pressure
            - Negative pressure on inhalation= increased heart rate
            - Positive pressure on exhalation= decreased heart rate
    - Common in larger or brachycephalic dogs
    - Can be eliminated with atropine

        - Lowers the parasympathetic nervous system activity
        - Increases firing of the SA node and conduction through the AV node
        - Opposes the action of the vagus nerve
- ECG “papers” has boxes- comes in 25 mm/sec or 50 mm/sec

<u>High Vagal Tone:</u>

- Vagus nerve= Cranial nerve X- controls parasympathetic nervous system to the heart, lungs, GI tract
- “High vagal tone”= Vagal tone is an internal biological process that represents the activity of the vagus nerve. Increasing your vagal tone activates the parasympathetic nervous system- this usually lowers heart rate and blood pressure

<u>Abnormal Heart Arrhythmias:</u>

- Sinus bradycardia
- Sinus tachycardia
- Impulse conduction problems
    - 1st degree, 2nd degree, 3rd degree, Bundle Branch Blocks
- Atrial fibrillation
- Atrial ectopic heartbeats
- Ventricular premature complexes (VPC’s)
- Ventricular fibrillation
- Asystole

<u>Impulse Conduction Problems:</u>

- Where the electrical impulse is blocked from proceeding in the normal fashion through the heart
- Examples: 1st degree AV block, 2nd degree AV block, 3rd degree AV block, Bundle Branch block
- Often requires a pacemaker

<u>AV Blocks:</u>

- 1st degree
    - Conduction through the AV node is delayed
    - Prolonged PR interval
- 2nd degree
    - Intermittent disruption of AV node conduction
- 3rd degree
    - No conduction through the AV node
- Bundle branch block
    - Blocked at the level of Left or Right bundle branches
    - Slower to depolarize one ventricle
    - QRS complexes are wider than normal

<u>Heart Rhythm- Abnormal:</u>

- **Supraventricular, ectopic heartbeat**
    - Depolarization above the ventricles
        - Atrial fibrillation, atrial premature beats
    - Atria stimulated to contract very quickly and differently from normal activity originating from the SA node
    - Ineffective and uncoordinated contraction of the atria
    - Causes irregular ventricle beats
    - P waves not associated with QRS complex

<u>Ventricular Arrhythmias:</u>

- Abnormal depolarization of the ventricles
    - Ventricular premature complexes
    - Extra heart beats that originate from the ventricles
- Ventricular fibrillation
    - Uncoordinated contraction of the ventricles
    - Ventricles quiver rather than contract properly
    - Leads to ventricular asystole

02/10/26

<u>To Sum Up Premature Complexes and Fibrillation:</u>

- Atrial
    - APC's
        - Beats originate in atria but not SA node
        - Out of sync with normal sinus rhythm
        - QRS complexes
    - Atrial tachycardia is 4 or more APC's in a row
    - Atrial fibrillation
        - Irregularly irregular
        - Supraventricular QRS complex
        - No P waves
        - Fibrillation "F" wave presents
- Ventricular
    - VPC's
        - Beats originate in ventricle before the SA node can fire
        - Out of sync with normal sinus rhythm
        - Wide and bizarre QRS complexes
        - No P wave
    - Ventricular tachycardia is 4 or more VPCs in a row with tachycardia
        - Doctor dependent
    - Ventricular fibrillation
        - No organized ventricular activity
        - No QRS complexes

<u>Blood Pressure:</u>

- Systole-contraction
- Diastole-relaxation
- Asystole-no contraction (flat line)
- Blood flows from areas of high pressure to low pressure
    - When the ventricles contract, blood flows from high-pressure ventricle to lower pressure arteries
    - Blood pressure is higher during ventricular systole
    - Higher on the arterial side of the circulatory system that the venous side
    - Lowest pressure is at vena cava's entrance to the right atrium (venous pressure or CVP)
    - Measured in mmHg

<u>What Effects Blood Pressure:</u>

- Resistance
    - A force that opposes blood flow
    - Smaller vessel=larger resistance
        - Therefore, vasoconstriction results in greater resistance, impeding blood flow
        - And vasodilation will allow blood to flow more freely (to a point)
- Cardiac output
    - HR, contractility, viscosity of blood

<u>Systolic and Diastolic Pressure:</u>

- **Systolic pressure**
    - Pressure of the blood in the arteries during ventricular systole
    - This is when arterial pressure is at its highest
- **Diastolic pressure**
    - Pressure in the arteries during ventricular diastole
    - Arterial pressure is at its lowest

<u>Mean Arterial Pressure:</u>

- **Mean Arterial Pressure (MAP)**
    - Average pressure of the blood in the arteries over the entire cardiac cycle. This is calculated or measured.
    - Calculation
        - MAP=DP + (⅓)(SP-DP)
    - Considered to be the perfusion pressure to the organs of the body
    - 60 mmHg is the lowest pressure able to sustain the organs
        - If the MAP falls significantly below this for too long, the organs will not get enough blood flow and will be ischemic

<u>Normal Blood Pressure Values:</u>

- Feline
    - Systolic: 120-140 mmHg

    - Diastolic: 50-80 mmHg
    - MAP: 60-100 mmHg
- Canine
    - Systolic: 130-160 mmHg
    - Diastolic: 50-80 mmHg
    - MAP: 60-100 mmHg

<u>What Affects Blood Pressure?</u>

- Cardiac output
- Blood volume
- Peripheral resistance
    - Increases with vasoconstriction
    - Decreases with vasodilation
- Blood viscosity
    - Viscous blood does not flow freely
    - Viscosity can change due to changes in the concentration of blood cells or plasma proteins
        - Hydration
- Increases in any of these factors will increase blood pressure
- Decreases will decrease blood pressure

<u>Homeostasis:</u>

- Blood pressure is maintained within normal limits as part of the body's homeostatic mechanisms
    - Pressure receptors respond to changes in arterial pressure
    - They send a signal to the cardiac and vasomotor centers of the brain
    - These centers send impulses via the autonomic nervous system to the heart and vessels
    - Changes occur in the cardiac output or blood vessel diameter in order to bring pressure back towards normal

<u>Hormonal Effects on Blood Pressure:</u>

- **Epinephrine and Norepinephrine**
    - Adrenal gland
    - Mimic the sympathetic branch of the autonomic nervous system (flight or fight response)
    - Increase blood pressure
- **Antidiuretic hormone (ADH)**
    - Hypothalamus → Pituitary → Adrenal gland
    - Reduces fluid loss by retaining H2O
    - Increases blood volume
    - Increases blood pressure
- **Angiotensin**
    - Liver → Adrenal gland

        - Causes vasoconstriction, which increases blood pressure
        - Stimulates secretion of the hormone aldosterone
            - Helps conserve sodium, increasing blood volume and blood pressure
- NAVEL- Naloxone, Atropine, Vasopressers, Epinephrine, Lidocaine

Indirect Blood Pressure Monitoring:

- Noninvasive
- Most common form of BP measurement
- Doppler- transducer is placed just proximal to the metacarpal/metatarsal pad and distal to the carpal pad (front limb on the palmer/plantar surface
- Oscillometric

Oscillometric:

- Systolic, diastolic, and MAP
- Automated, less technical skill
- Less accurate
    - Patients under 5 kg, arrhythmias, tachycardia or bradycardia, vasoconstriction, hypothermia, moving or shivering
- Choose appropriate cuff size
- Place proximal to artery

Doppler:

- Systolic only
- Affordable and readily available
- Suitable for patients with hypotension and arrhythmias
- Appropriate BP cuff proximal to artery
    - Width 30-50% of circumferences at site of placement
- Clip hair over appropriate artery
    - Dorsal metatarsal, ventral digital, tail
- Position directly over artery
    - Listen for clear, loud pulse
- Sphygomomanometer attached to cuff
    - Inflate cuff, slowly deflate until pulse becomes audible
- Take multiple measurements. Note location and size of cuff in medical record
- Hypertension
    - Cause
        - Stress, fear, overhydration, metabolic disease (renal, endo, etc)
    - Signs
        - Increased CRT, injected or brick red mm, bounding pulses, gradual or sudden blindness
- Hypotension
    - Cause
        - Hypovolemia, vasodilation, myocardial depression, arrhythmias, vagal stimulation

02/12/26

Pulse Oximetry (SPO2):

- Measures the amount of oxygen saturation of red blood cells
    - Emits and reabsorbs red and infrared light
        - Protects from light/ it may interfere with results
    - Normal values above 95%
        - Below 90%=hypoxia
    - Locations
        - Tongue, lip, prepuce, vulva, digit, etc
            - Hypotension could affect results
        - Positional. +/- moisture
    - Anesthesia, emergency

02/18/26

IV Catheter Basics:

Why Do We Place IV Catheters?

- Purpose: provide vascular access
    - Usually need temporary venous access for medications, fluids, and electrolyte replacement therapy
    - Can be used for transfusion of blood products
    - Emergency IV access

Where Do You Place an IVC?

- Mostly use the common peripheral venipuncture veins
- Do not use jugular unless placing a jugular catheter
- Try to use an area that is easiest to take in
    - Always start as distally as possible
- Veins for Canine IV Catheterization
    - Cephalic
    - External jugular
    - Lateral saphenous

Considerations When Choosing an IVC:

- Need to have sufficient length and secure it properly
- Choose a large diameter in case rapid fluid administration is necessary
- Be sure to use an administration set with an injection port so medication can be administered
- Take care to not occlude the catheter when securing the patient to the surgical table (if applicable)
- BP cuff placement

Types of Catheters:

- Common types of IVC:

    - Through-the-needle
    - Over-the-needle
    - Winged needle (butterfly)
    - Multi-lumen catheter

Over the Needle:

- Most common type of catheter used today
- Catheter is fitted outside or over a steel needle
- The needle tip extends past the catheter about a millimeter
- Easy to place
- Less traumatic
- Relatively inexpensive
- 1-2 days

Through the Needle:

- Used most commonly in critical care patients
- Used for central lines- jugular vein or caudal vena cava
    - Strict aseptic technique
- Passed through or inside the needle
- Are longer than over the needle catheters and have a range in diameters
    - Hypertonic solutions
- Complex
- Expensive
- Time consuming
- 4-5 days with proper care
- Winged needle (butterfly)
    - Short term use- needles have plastic wings that facilitate placing or taping in
    - Can lacerate the vein
- Multi-lumen catheter
    - Two to three separate lumens in one catheter- allow for simultaneous infusions in a single catheter site
    - Wire-guided

Catheter Placement:

- Common catheter sites include the Cephalic, Saphenous (medial-cat, lateral-dog)
- Wash your hands or wear gloves
- Shave the site- must use 40 blade minimum
- Aseptically clean the site with antiseptic scrub (3 scrubs)
- Can use a facilitative incision or relief hole to reduce skin tension and friction against the catheter
- Start distal on the limb in case you need to try repeat attempt more proximal
- Aseptic introduction is critical to avoid contamination and possible infection
- Never reinsert the needle if you fail to thread the catheter. Major problems can arise
- 20-30 degree angle

- Brisk motion
    - Too slow- pushes the vein out of the way
    - Too fast- puncture through the vein
- WATCH FOR FLASH

<u>Preparation- Tape Selection:</u>

- Porous tape
    - Length- 2-3 times the diameter of the limb
- ½ inch anchor tape
- 1 inch “drain board” with tear
- 1 inch for T-port
- ½ x 1 ½= chevron (bra strap)

<u>Catheter Taping:</u>

- Anchor is very important for security. Usually secured with ½ tape sticky side up
- “Drain board” sticky side down
- Chevron tape to secure t-port, sticky side up, but with a twist
- Fluid line securing tape, sticky side down

<u>Catheter Maintenance:</u>

- Changing the catheter every 48-72 hours= old thinking
- Unwrap and check puncture site at least every 24 hours, best to do every 12 hours
- Heparin lock consider if not using the catheter to prevent clotting
    - Heparin saline recipe: 250-500 IU heparin to 250 ml normal saline
    - Alternate technique: can simply flush with normal saline more frequently

<u>Complications- Catheter:</u>

- Occlusion
    - Clot formation
    - Kink
- Malpositioning
    - Location of the vessels is unknown
    - Taping of the catheter is not optimal
    - Movement
- Burning or fraying of the catheter
- Breaking the catheter
- Embolism

<u>Complications- Patient:</u>

- Thrombophelbitis
- Phlebitis
- Extravasation
- Thrombus formation
- Infiltration of subcutaneous tissue

- Hemorrhage
- Subcutaneous hematoma formation
- Systemic problems may involve pyrogenic reaction, circulatory overload, air embolism, or allergic reaction

A Note About IV Catheters for Chemotherapy:

- IV catheters for cytotoxic solution
    - Must be a “clean stick”
- Perform frequent patency checks
    - Watch for signs of extravasation
    - Correction of extravasation

02/24/26

Assessing Dehydration Fluid Therapy:

The Goal of Fluid Therapy:

- Rehydration
- Replacement
- Resuscitation

What Are the Indications For Fluid Therapy?

- Maintaining hydration
- Replacing fluid deficit (dehydration)
- Replacing ongoing losses
- Treating decreased oncotic pressure
- Treating hypovolemia
- Treating shock states
- Improving or increasing urine production
- Correcting acid/base balance or electrolyte disorders
- Maintaining intravenous access and delivering other medications

Body Fluids:

- Fluid compartments
    - Intracellular fluid- ICF
        - Fluid within all of the cells of the body
    - Extracellular fluid- ECF
        - Interstitial fluid- fluid between the cells of tissue
        - Intravascular fluid= plasma- the liquid component of blood and lymph in which erythrocytes, leukocytes, and platelets are suspended
- The adult animal’s body is 55-65% water by weight
- Young animal’s body is up to 85% water by weight
- What are the barriers between compartments?
    - ICF/ECF barrier= the cell membrane
    - Interstitial/intravascular barrier= vascular endothelium and glycocalyx (protein/lipid layer on endothelial surface)

Water Balance:

- Natural water losses
    - Sensible (measureable)
        - Urination
    - Insensible (not measurable)
        - Defecation
        - Respiration
    - Contemporary
        - Diarrhea
        - Vomiting
        - Hemorrhage
        - Sialorrhea- excessive tear production
        - Hyperhidrosis
- Circulating blood volume
    - Dogs= 80-90 ml/kg
    - Cats= 40-60 ml/kg
    - Horses= 80 ml/kg
- Also, pyrexia will cause the body to burn its water faster (fever)
- Dehydration: Water loss > water gain
    - Decreased blood volume
    - Decreased blood pressure
    - Reduced tissue perfusion
- Electrolyte loss
    - Usually results in metabolic acidosis
- 10% loss of body water= serious metabolic disturbances
- 30% loss of body water= death

- Obtaining a History:
  No food or water intake for the past 24 hours
- Presence of vomiting, diarrhea, or increased volume or frequency of urination
- What questions would you ask?

Intravascular Loss:

- Hypotension
- Tachycardia
- Obtundation
- Hypothermia
- Cool extremities
- Prolonged CRT
- Bounding/thread pulses
- Weight loss
- BUT WHY???

Interstitial Loss:

- Skin turgor
    - Always use the same spot
- CRT
- Weight
- Heart rate/ pulses
- Mucous membranes
- Sunken eyes (enophthalmos)
- Elevated PCV/TP

EVALUATING DEHYDRATION

ESTIMATED % DEHYDRATION	PHYSICAL EXAMINATION FINDING
<5%	• Not detectable
5–6%	• Some change in skin turgor
6–8%	• Mild decreased skin turgor • Dry mucous membranes*
8–10%	• Obvious decreased skin turgor • Retracted globes within orbits
10–12%	• Persistent skin tent due to complete loss of skin elasticity • Dull corneas** • Evidence of hypovolemia
>12%	• Hypovolemic shock • Death

Nursing Assessments:

- Altered electrolyte/chemistry status
- Altered erythrocyte status
- Altered gas exchange
- Altered mental status
- Altered urination
- Fluid volume deficit
- Impaired self care
- Impaired thermoregulation
- Reduced cardiac output
- Reduced perfusion

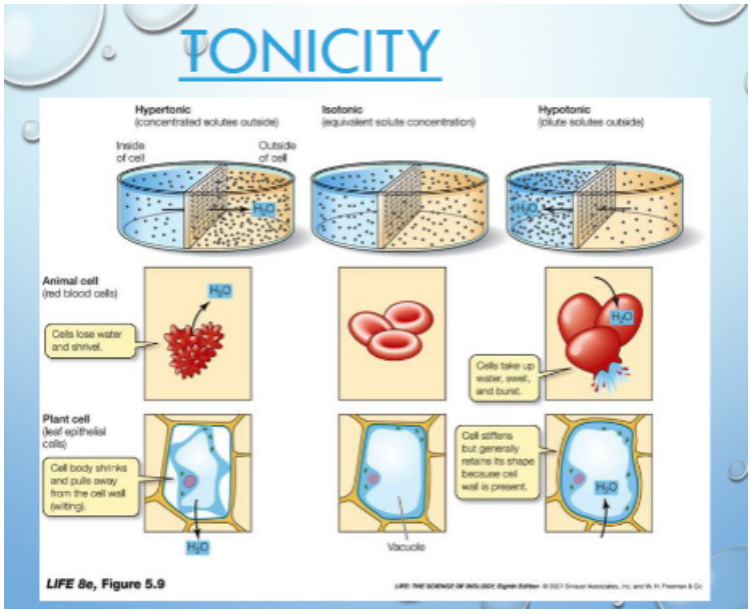
Packed Cell Volume and Total Protein/Solid:

- If water intake is reduced or water excretion increased and, there is a net loss of body water
- Dehydration= loss of body water concentrates the cells and proteins suspended in the blood- high PCV= polycythemia
- Therefore, PCV and TP can be used to assess hydration status

- Per Idexx normals are:
    - Dog
        - 37-55%
        - 6-8 g/dL
    - Cat
        - 24-55%
        - 5.7-8.9 g/dL
- Note, other sources may vary
- Maintain blood volume
- Replacement of water losses due to illness or injury
- Prevent dehydration
- Correct acid/base (blood pH) disturbances
- Correct or prevent electrolyte imbalances
    - Na+, K+, Cl+, etc, from the body

<u>Fluid Therapy:</u>

- Keys to fluid therapy planning
    - Right type
    - Right amount
    - Right speed
- Considerations
    - Body fluid compartments
    - Fluid types
    - Rates and routes
    - Effectiveness monitoring
    - Potential complications



We Need to Take The Patients Condition Into Consideration:

- If a patient is dehydrated what would happen if we give them
    - Hypotonic solution
- If a patient is overhydrated what would happen if we give them
    - Hypertonic solution
- If a patient is appropriately hydrated what would happen if we give them
    - Isotonic solution

Diffusion Ability:

- Crystalloids
    - Solutions containing substances that pass easily through cell membrane
        - Electrolytes
        - Always isotonic or hypotonic with one exception (hypertonic saline)
- Colloids
    - Solutions that contain large molecules

Common Crystalloids:

- LRS: Na+, Ca 2+, K+, lactate
    - Most commonly used
- Maintenance of normal hydration
- Correction of dehydration
- Replacement of commonly depleted electrolytes (vomiting, diarrhea, renal disease)
- Tx of shock
- Correction of metabolic acidosis
    - Healthy livers will convert lactate into bicarbonate
- Normosol-R, Multisol-R, Plasma-Lyte:
    - Na+, Ca 2+, Cl-, K+, MG
        - Similar to LRS with similar uses, preferred with patients with liver disease because acetate and gluconate are converted to pH buffers in the muscle rather than the liver
- Saline: Na+, Cl-
    - Sodium is the major solute in extracellular space
    - Sodium based solutions rapidly distribute, and increase the volume of both plasma and interstitial fluid
    - Use for patients with hyponatremia (low sodium) and hyperkalemia (high potassium)
- Dextrose: Glucose
    - Used to treat hypoglycemia, especially neonates, 5% is isotonic when administered, but once glucose is metabolized, the remaining water is hypotonic. Can deplete electrolytes in the body, due to excretion of water through the kidneys.

02/26/25

Collids:

- **Hypertonic** solutions that contain large molecules
    - Cannot pass through cell membrane
    - Referred to as “volume expanders”
    - Colloids remain in plasma and serve as osmotically active particles
        - Colloids will hold fluid in the intravascular space
        - Draw fluid from the interstitial space into the vascular space to increase plasma volume
        - Colloids are 5-10 times more effective than crystalloids at plasma volume
- Albumin is an osmotically active protein commonly found in blood. Colloids serve as a replacement for albumin to treat hypoproteinemia and associated symptoms (ascites, edema)
- Colloids may be used alone or in combination with crystalloids
    - Treat hypovolemia resulting from blood loss

<u>Common Colloids:</u>

- Plasma
- Whole blood
- Packed red blood cells
- Hydroxethyl starches derived from corn
    - Will remain in plasma for 12-48 hours
    - Does not contain RBC and clotting factors
- Dextrans
    - 4-8 hours

<u>Selection of Fluids:</u>

- Administering colloids with crystalloids reduces the need for crystalloids 40-60%
    - Fluid overload less likely
    - Resuscitation times are shorter
- Many conditions can cause increased capillary permeability
    - Parvo, severe, GI disease, pancreatitis, septic shock, massive trauma, heat stroke, and others
    - Hetastarch or stroma-free hemoglobin are the choice for resuscitation

<u>Routes of Administration:</u>

- Oral- PO
    - Natural intake preferred for fluid maintenance
    - CAN NOT be used with patients:
        - Vomiting
        - Malabsorption
        - GI obstruction
        - Or rapid IV fluid volume is needed (shock)
    - Commercially available electrolyte solution are preferred

<u>Enteral Administration:</u>

- Enteral
    - Gavage tubes and needles are commonly used with rodents, reptiles, birds
    - Nasogastric tubes can be used in rabbits
    - Esophagastomy tubes are surgically placed and can be used long term

03/05/26

What Can We Add to the Fluid Bag?

- KCL
- Antibiotics
- Pain medications
- Antiemetics
- Sedatives
- Vitamins
- Dextrose
- Sodium Bicarbonate

Additives:

- Always label bags with:
    - Date opened
    - Types of additives
    - Amount of additives
    - Your initials
- Fluids that are not contaminated or do not contain additives can be used as SQ fluids
    - Clinics will vary as to what is an acceptable amount of time to keep

Storage of Fluids:

- Crystalloids and synthetic colloids should be stored at room temp, and away from sunlight, WARM BEFORE USE
- First in=first out, check expiration dates
- Discard fluids that are no longer sterile
    - Opened but not used
    - Contains additives
    - Electrolytes/additives precipitate or become cloudy

Maintenance Phase:

- Provide fluid support to maintain current condition
    - After dehydration is corrected, and there are no more ongoing losses
    - Isotonic crystalloids are often used for maintenance fluid therapy
    - Patients with normal cardiac and renal function can easily process the excess sodium and electrolytes in the isotonic fluids
- Animals need 50-60 mLs H2O/KG/24 hrs or 25-30 mLs/#/day
    - 60 ml/kg/day

Replacement Phase:

- Correction of dehydration, replacement of ongoing losses, and provision of maintenance fluid requirements
    - Isotonic crystalloids most common
        - Isotonic in nature (vomiting, diarrhea, blood loss)

Fluid Overload:

- Heart failure, kidney failure, technician error
    - Pulmonary edema
        - Left sided heart failure
    - Tachypnea, dyspnea
        - Respiratory rate and pattern monitored
        - Increase of more than 20%
    - Nasal d/c
    - Cough
    - Crackles
    - Increase in body
    - Weight increase
- Cavity effusion or peripheral edema
- Vomiting
- Salivation
- Restlessness
- Chemosis
- Hypertension
- Increased PCV
- Polyurea

03/10/26

Ophthalmology

Common Ophthalmic Exam Tests:

- Fluorescein stain
- Menace response
- Palpebral reflex
- Pupillary light reflex (PLR(
- Schirmer tear test (STT)
- Tonometer

Before You Start an Ophthalmic Exam, Assess if the Patient is Visual…

- Can the patient see?
    - How to assess?
        - Navigation of room
        - Throw cotton balls
        - Menace response
            - This is not a reflex! It is a learned response. Very young puppies and kittens will not have this response.

- Any other reason the patient cannot blink?

Examination of the Eye:

- Redness, swelling, discharge
  - Note symmetry and character
- Eyelids/eyelashes
  - Opening normally? Masses? Ectopic cilia? Entropion (lower eyelid rolling in toward the eye)? Ectropion?
- Symmetrical pupils?
  - Terms: Anisocoria, Strabismus, Nystagmus
- Enlarged or protruding eyes
  - Glaucoma, mass behind the eye
  - Can you retrobulbar press the globe?
- Exopthalmos
- Endophthalmos

Ocular Reflexes:

- Corneal reflex
- Palpebral reflex
  - If too deep under anesthesia, this reflex may be lost
- Menace reflex
- Doll's eye Reflex
- Nystagmus
  - Caused by improper function of a nerve
  - Can be genetic (common is Siamese and Himalayan cats)
- Dropping cotton balls or other light material in front of their face to see if they watch it fall
  - Cannot use items that make noise- will not only test sight at that point= cheating!

Schirmer Tear Test:

- Measures the amount of tears produced
- Perform prior to other tests
- Normal values= 15-20 mm

Fluorescein Dye:

- Checks for corneal ulcers
- Checks the patency of the lacrimal system
- Wood's lamp or cobalt filter may be used to illuminate the dye and make reading easier
- Supplies
  - Stain
  - Eye wash
  - Cotton or gauze
  - Sterile saline
  - Black light
  - Proparacaine

        - Good restraint- what do you want to be mindful of?

Ophthalmoscope:

- Used to directly examine the internal structures of the eye. Magnification= 15X
- Green filter- provides better contrast of red blood vessels
- Slit lamp= helps determine “cloudiness” in anterior chamber
- UV light= helps highlight fluorescein stain

Ophthalmic Instruments:

- Indirect Ophthalmoscopy
    - Better view of the retina
- Transilluminator
    - Bright, focused light may have cobalt blue cap
- Schiotz or Pen Tonometer
    - Measures intraocular pressure of the eye

Tonopen:

- To measure intraocular pressure (IOP)
- Items needed for this test:
    - Tonopen
        - May need to calibrate machine first
    - Tonopen cover
    - Proparacaine
    - Good restraint
    - Need to know how to read results
- Glaucoma is suggested at a reading over 25 mmHg
    - IOP normals:
        - Canine= 16-25 mmHg
        - Feline= 14-25 mmHg

Eyelid Abnormalities:

- Entropion
    - Turning inward of the eyelid- can cause corneal ulcers
    - Breeds prone= St. Benard, Sharpei, Chow Chow
- Ectropion
    - Turning outward of the eyelid- causes conjunctivitis
    - Breeds prone= bassets, St. Bernards, Boxers
- Usually genetic or congenital
- Tx: Surgery if causes problems

Distichiasis:

- Cilia originate from abnormally located follicles just posterior to the normal cilia
- Irritate the cornea causing ulceration
- Cause: genetic or congenital

- Treatment:
    - Surgical excision
    - Cryo surgery
    - Plucking hairs

<u>Blepharitis:</u>

- Inflammation of the eyelid
- Causes:
    - Trauma
    - Bacteria- common in dogs and cats
    - Fungus- common in horses
    - Allergies (seen after vaccinations, bee stings, etc)
    - Parasites- Thelazia sp
- Treatment:
    - Anti-inflammatory drugs combined with antimicrobial or antifungal agents

<u>Epiphoria:</u>

- Excessive tearing
- Results in staining of the medial canthus hairs- is more pronounced in lighter coated dogs
- May cause dermatitis with inflammation
- Cause:
    - Genetic
    - Increased production- irritation
    - Block drainage- puncta or nasolacrimal duct
- Treatment:
    - Flushing the puncta and nasolacrimal duct
    - Removing inciting cause

<u>Neoplasia:</u>

- Squamous Cell Carcinoma- malignant
    - Eyes, ears, nose
    - Called “Cancer Eye” in large animals
- Treatment:
    - Usually requires surgical removal
    - Benign or malignant
    - Causing disturbance of function

<u>Cherry Eye:</u>

- Protrusion of the 3rd eyelid or its gland
- Results in abrasion and drying of the gland and possible damage to the cornea
- May result is KCS (Keratoconjunctivitis Sicca)
- Cause:
    - Genetic, traumatic, or allergic

- Treatment:
    - Surgical replacement
    - May have to be removed in severe cases

Foreign Bodies:

- Can be any type of foreign material in the eye
- Usually associated with trauma
- Commonly causes ulcerations
- Normally seen with the naked eye unless hiding under the 3rd eyelid!

Proptosed Eye:

- A medical emergency, especially if you want to save vision
- Must anesthetize patient and replace eye into the socket
- Study suggest 60% of canine and 100% of felines will lose vision in that eye

Lens Abnormalities:

- Loss of transparency of the lens due to the buildup of a protein material
- Nuclear sclerosis is an age related change that does not usually affect vision
- Cataracts block light passage and decrease vision
    - Location of on the lens- capsular or subcapsular
    - Age on onset- congenital, developmental or juvenile, senile, or senescent
    - Cause- traumatic, genetic
    - Degree of Maturation

Horner's Syndrome:

- Combination of four clinical signs:
    - Ptosis (drooping of upper eyelid)
    - Miosis (pupil constriction)
    - Elevated third eyelid
    - Enophthalmos (globe sunk back into socket)
- Often acute and unilateral
- Caused by denervation of sympathetic pathways
- Multiple etiologies (trauma, neoplasia, infection, others)

Retinal Detachment:

- A separation of the retina from the choroid
- May be congenital or due to trauma, infectious diseases, hypertension, or neoplasia
- Signs:
    - Loss of vision, appearance of retinal vessels through the pupil, increased blood pressure
- Treatment:
    - Diuretics to reduce edema, antihypertensive meds, antibiotics, and corticosteroids
    - Surgery may be needed

Retinal Hemorrhage:

- Bleeding into retina
    - Blood vessels of retina or choroid
    - Congenital deformities (Collie eye anomaly)
    - Circulatory disorders
    - Clotting abnormalities
    - infection/inflammation
    - Neoplasia
    - Trauma
- Signs:
    - Mild
        - No obvious signs
    - Severe
        - Decreased vision, dilated pupils
- Treatment:
    - Treat underlying condition

Conjunctivitis:

- Inflammation of the conjunctiva
- Causes:
    - Allergic infectious- Canine Distemper or Chlamydia
    - Immune mediated
- Signs:
    - Pruritus and alopecia of surrounding area
    - Increased tear production and pain
- Treatment
    - Eliminate cause or minimize inflammation
        - Topical steroids

Keratoconjuntivitis Sicca:

- Decrease in tear production
- Drying of the cornea and a mucoid film over the eye
- Brachycephalic
- Common in older dogs and may be caused by drugs (e.g. Sulfa drugs) or immune mediated
- Dx: Less than 10 mm on a Schirmer Tear Test
- Treatment:
    - Medical- topical cyclosporine (optimmune)
    - Surgical- Parotid Duct Transplant

Corneal Edema/Ulceration:

- Signs:

        - Edema, vascularization, scar formation, pigment deposition, pain, blepharospasm, and cellular infiltration
- Causes:
    - Trauma (incl foreign bodies), infections (bacterial, fungal viral-ICH, Herpes), chronic glaucoma, radiation, drugs, burns, and hypotonic solutions
- Treatments:
    - Topical antibiotics and mydriatics (dilate the pupil)
    - May require surgery if severe or non-healing
    - NEVER use corticosteroids
        - Interfere with healing
        - Will make ulcer worse

03/12/26

Glaucoma:

- Increase in the ocular pressure
    - Increased production of aqueous humour
    - Decreased drainage- due to blockage
- Signs
    - Corneal edema, enlarged vessels, dilated pupil, vision loss, swelling of the eye (buphthalmos), ocular pain, loss of corneal sensitivity
- Treatment
    - Long-term administration of drugs
    - Diuretics to remove excess fluids
    - Miotics (Pilocarpine)- constrict pupil so iris doesn't block passage of aqueous humour from ciliary body to anterior chamber
        - Prevents buildup of aqueous humor in the posterior chamber

Uveitis:

- An inflammation of the middle layer of the eye (uvea)
- The most common type is an inflammation of the iris called iritis (anterior uveitis)
- Symptoms include redness, pain, light sensitivity, blurred vision, and dark floating spots in the field of vision
- Look for flare with slit lamp

Common Surgical Procedures:

- Conjunctival or 3rd Eyelid Flap
    - A flap of conjunctiva is sutured over the globe to provide a temporary covering for the eye.
    - Used in corneal ulcers with prolonged healing
    - Buttons or tubing may be used with tension sutures to prevent tearing of the thin conjunctiva.
- Enucleation
    - Removal of the entire globe

        - An intra-ocular prosthesis may be used to maintain more normal appearance of the eye

<u>Ocular Administration Routes:</u>

- Topical- most common route
    - Make sure tip does not come in contact with the eye
        - Transfers microbials, idiopathic conditions, scratch the cornea
    - Should be applied no less that 3-4 times a day
    - Drops
        - Try to hit center of the eye
        - Penetrate faster than ointments
- Ointments
    - Place on lower conjunctiva
    - Stays in eye longer and do not tear out as easily

<u>Ocular Medications:</u>

- Antimicrobials
- Miotics
    - Pupil constriction
    - Glaucoma
- Mydriatics
    - Pupil dilation
- Anti-inflammatory steroids